RAG Assignment: Baseline Retrieval vs. Cross-Encoder Re-Ranking

Hritik Ranjan (Roll No. 2025202031)

October 19, 2026

code: https://github.com/HRITIK-RANJAN/RAG-BASELINE-VS-RERANKING

Setup & Environment

- **Dataset:** *Netrunner Comprehensive Rules v26.03*, a 99-page rulebook with many restrictions and exceptions.
- **Models:** `all-MiniLM-L6-v2` embeddings, FAISS cosine search, `ms-marco-MiniLM-L-6-v2` re-ranking, and Gemini Flash Lite for answers (temperature 0).
- **Corpus and machine:** 602 structure-aware chunks were used. Both pipelines use the same frozen corpus.

1 Two-Stage Retrieval

1.1 Pipeline 1: baseline retrieval

The bi-encoder embeds the query and all chunks independently. FAISS returns the top 10 by cosine similarity, and the top 3 are sent to the LLM.

1.2 Pipeline 2: advanced re-ranking

The second pipeline re-scores those same 10 candidates with the Cross-Encoder, then sends its top 3 to the identical LLM prompt.

2 Architecture comparison

A **Bi-Encoder** creates independent embeddings for the query and each chunk, then compares them with cosine similarity. Chunk vectors can be computed offline, making it fast, but word relationships such as "cannot" or "only if" may be blurred.

A **Cross-Encoder** reads the query and chunk together with joint self-attention across their tokens and gives one pairwise score. This captures conditions and wording better, but requires a separate model pass for every candidate at query time.

3 Failure Mode Analysis (Stress Testing)

Query set. The final set contains 23 stress-test queries covering controls, keyword traps, exceptions, negation and buried answers. It is not a random sample.

	Hit@3	Recall@3	MRR@10	Mean gold rank (in top 10)
Pipeline 1 (baseline)	0.652	0.514	0.637	2.29
Pipeline 2 (re-ranked)	**0.696**	**0.638**	**0.687**	**2.24**

Table 1: Retrieval metrics on the 23 final queries (essential gold chunks only).

By category (Hit@3 baseline → re-ranked): buried 0.40 → 0.60, control 0.83 → 1.00, exception 0.67 → 0.67, keyword trap 0.60 → 0.60, negation 0.75 → 0.50. Re-ranking moved the best gold chunk up in 5 queries, down in 6 and left it in place in 12. The two models agree only moderately on the order of the same 10 chunks (median Spearman 0.38, range −0.20 to 0.94).

Examples. Q29 is a keyword trap: "flip face up" refers to the rule term "rez". Its relevant chunk moved from rank 4 to rank 1. Q02 and two buried-answer queries also moved into the top 3.

Exceptions and negation. Q18's "cannot" rule (1.2.2) stays at rank 6 under both two-stage models; when the cross-encoder scores all 602 chunks, it ranks only 20th. For Q12, the gold chunk starts at bi-encoder rank 20, outside the top 10 candidates, so the two-stage re-ranker cannot see it.

Limitations. Re-ranking also hurt: Q36 moved from rank 1 to rank 9. Q23 has two essential passages: 3.8.5 fell from rank 1 to rank 7, while 10.3.1d rose from rank 8 to rank 4—still outside the top 3. This illustrates that the best-gold metric alone can hide a harmful move of another required passage.

ID	Category	Gold chunk: bi → CE rank	Query
Fixed by re-ranking			
Q02	control	5.2.6: #7 → #1	What basic actions can the Corp always take during their turn?
Q29	keyword trap	8.1.2e: #4 → #1	How does the Corp flip an installed card face up so that it starts working?
Q35	buried	7.4.7: #4 → #1	In what order are cards from the Corp's deck offered to the Runner when R&D is breached?
Q41	buried	7.3.8: #4 → #1	If stealing an agenda would cause a breach of the same server during an ongoing breach, when does the second breach happen?
Hurt by re-ranking			
Q23	negation	3.8.5: #1 → #7; 10.3.1d: #8 → #4	Can the Runner have two consoles installed at the same time?
Q32	keyword trap	4.2.7: #3 → #4	What happens when the Corp runs out of cards in their deck and has to draw?
Q36	buried	7.5: #1 → #9	What does the Runner do in a breach when there are no more cards left to choose from?
Failing in both / not in the top 10			
Q18	negation	1.2.2: #6 → #6	If one effect says a player must do something and another says they cannot, which applies?
Q34	buried	7.4.6: #4 → #5	While the Runner is breaching HQ, a new card enters the Corp's hand. Can the Runner access it?
Q12	exception	1.13.13: not in top 10	What happens to cards hosted on a card that moves from the Corp's to the Runner's score area?
Q42	keyword trap	10.1.1: not in top 10	What happens if I have two copies of a card with the diamond symbol in play?

Table 2: Rank shifts of the gold chunk for every final query whose outcome is not "both ok".

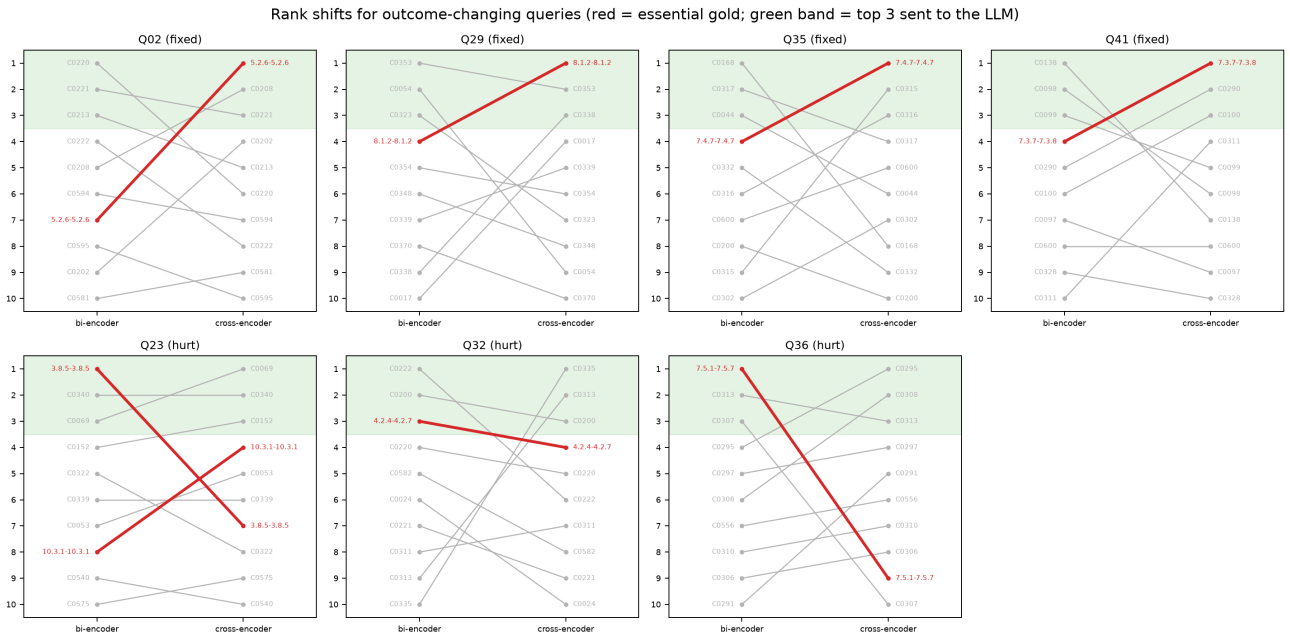


Figure 1: Rank shifts for the four fixed and three hurt queries. Red = essential gold chunk; green band = top 3 sent to the LLM.

3.1 Answer 1 vs. Answer 2

Answer 1 had 15 correct answers, 7 refusals and 1 hallucination. Answer 2 had 16 correct answers, 6 refusals and 1 hallucination. Four baseline refusals became correct after the relevant chunk entered the top 3; three correct baseline answers became refusals when re-ranking moved that chunk out. There were no 'omits clause' or 'misstates' cases: the refusal instruction turned missing evidence into refusals. Q34 is the only hallucination: its expected conclusion is indeed "yes", but both answers are labelled hallucinated because none of their supplied passages supports that claim—rule 7.4.6 was rank 4 for Pipeline 1 and rank 5 for Pipeline 2. A concrete improvement is Q29: Answer 1 refused, whereas Answer 2 used the retrieved rule to explain rezzing correctly.

4 Latency

Timings exclude model loading. Retrieval stages use 20 repeats over the 23 queries; the LLM uses one real call per query and pipeline, timed separately because API latency is variable.

	Pipeline 1 (ms)			Pipeline 2 (ms)		
Stage	median	mean	p95	median	mean	p95
Query encoding (bi-encoder)	4.12	4.23	4.72	4.12	4.23	4.72
FAISS search, top 10	0.02	0.02	0.03	0.02	0.02	0.03
Cross-encoder, 10 pairs	–	–	–	35.04	34.64	36.14
Retrieval subtotal	**4.15**	4.25	4.75	**39.22**	38.89	40.61
LLM generation (Gemini)	1322.5	5036.4	7439.2	1764.6	5418.8	14826.3
End-to-end	1326.6	5040.6	7443.3	1802.3	5457.6	14865.3

Table 3: Latency per query (Apple M4, MPS).

Re-ranking itself adds about 35 ms per query. The LLM median difference (1.32 s for Pipeline 1 versus 1.76 s for Pipeline 2) is API noise, not a re-ranking cost: the prompts were sent independently at different times. The means near 5 s are driven by one 61.5 s Pipeline 1 call (Q31) and one 31.4 s Pipeline 2 call (Q28). Thus the 1.3 s versus 1.8 s end-to-end medians should not be read as the price of re-ranking; the measured local cost is 35 ms.

5 Why not cross-encode the whole corpus?

The Cross-Encoder cannot precompute a score because it needs the query and chunk together. Its cost therefore grows linearly with the number of chunks: about **2.74 ms per pair**. The Bi-Encoder first narrows the collection cheaply, leaving only 10 pairs for accurate re-ranking.

Chunks N	Cross-Encoder on all N	Two-stage (encode + exact search + CE on 10)
602	1.66 s	0.04 s
100,000	274 s (4.6 min)	0.04 s
1,000,000	2,736 s (46 min)	0.07 s
10,000,000	27,364 s (7.6 h)	0.37 s

Table 4: Time per query (above 200k chunks: linear extrapolation; approximate indexes would make the two-stage column grow even more slowly).

For this corpus, re-ranking all 602 chunks takes about 1.66 s (the measured median), compared with about 0.04 s for two-stage retrieval. This also measures the recall ceiling: full-corpus cross-encoding puts Q12 at rank 1 and raises Hit@3 to 0.739, versus 0.696 for the two-stage pipeline. At 1 million chunks, the same approach would take roughly 46 minutes per query. The quality gain comes at about 1.62 s extra per query on this corpus.

Limitations

The query set was selected for stress testing, and timings are from one machine. LLM latency is noisy; scaling figures beyond the measured corpus are extrapolations.